

DIPIN KUNHAMBU NAIR

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SUMMARY

Software engineer with 7+ years of experience building production-level Python and C++ software for simulation, automation, and developer tooling. Maintains PyMechanical and contributes to the PyAnsys open-source ecosystem, with ownership of testing, releases, CI/CD, and documentation. Seeking software roles focused on scalable Python platforms, open-source tooling, and reliable customer-facing software.

EDUCATION

University of Cincinnati

Master of Science in Mechanical Engineering | GPA: 4.0/4.0

January 2021 – August 2023

Cincinnati, Ohio

Indian Institute of Technology, Hyderabad

Bachelor of Technology in Mechanical Engineering | GPA: 8.24/10

August 2012 – August 2016

Hyderabad, India

SKILLS

Programming: Python, C++, C#, MATLAB

Software Engineering: Unit Testing, Object-Oriented Programming, Automation, Agile/Scrum

Tools/DevOps: Git, Docker, Linux, GitHub Actions, Azure DevOps, CI/CD

PROFESSIONAL EXPERIENCE

Synopsys Inc.

Senior Software Engineer

August 2023 – Present

Austin, Texas

- Architected and maintained PyMechanical, a PyAnsys open-source Python library enabling programmatic access to Ansys Mechanical.
- Engineered Docker-based containerization and Azure DevOps CI/CD pipelines, reducing release cycles by 30% across Windows and Linux platforms.
- Developed unit and integration testing frameworks, increasing automated regression coverage from 70% to 90%.
- Resolved critical production bugs in ANSYS Mechanical core software (C++/C#) and improved reliability of customer-facing workflows.

Ansys Inc.

Software Developer Intern

January 2023 – April 2023

Pittsburgh, Pennsylvania

- Contributed ANSYS Mechanical release features with the Linear Dynamics team, supporting production C++/C# development workflows.
- Validated changes through Windows/Linux developer builds, Azure pipelines, and ARM tests before integration.
- Applied COM principles to extend Mechanical interfaces; partnered with test teams to triage defects and document release functionality.

Procter & Gamble

Robotics Research Assistant

August 2021 – August 2022

Cincinnati, Ohio

- Automated fragrance consumption analysis for 600+ air fresheners using robotic batch testing workflows.
- Enabled robot weighing, QR-code computer vision, and validated data uploads to Siemens server workflows.
- Created a Python PyQt GUI to improve customer interaction and standardize test execution.

Tata Consultancy Ltd

Systems Engineer

October 2019 – December 2020

Bangalore, India

- Earned Design for Six Sigma Black Belt certification by supervising 2 projects as team leader and facilitator.
- Coordinated development of 12 efficiency modules in MASTA software using C++.
- Led a cross-functional team of 4 to automate meshing in SimLab using JavaScript for simulation analysis.

General Motors

Senior Engineer

September 2016 – September 2019

Bangalore, India

- Automated Abaqus system modeling with VBA scripting, reducing manual effort by 85%.
- Earned 7 recognitions, including Work of Wonder and Pat on the Back, for transmission efficiency optimization.